Techno-economic feasibility study of a hybrid Biomass/PV/Diesel/Battery system for powering the village of Imlil in High Atlas of Morocco

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***Abstract*— In this study, we propose a techno-economic feasibility analysis to meet the energy needs of a rural village in the High Atlas Mountains in Morocco. We use different hybrid systems based on wood chips and PV modules. The electrical load data were taken into account for the heat and electricity needs of the village. Homer (Hybrid Optimization Model for Renewable Energy) was chosen to conduct our techno-economic feasibility analysis using biomass potential and solar irradiance data from this area. All possible scenarios were obtained as a solution by the HOMER analysis, specifically the total net current cost (NPC), the cost of electricity (COE) and wood consumption were obtained as a solution by this analysis.Keywords; CHP; PV; Diesel; Battery; High Atlas; Wood chips**

I. INTRODUCTION:

The issues of limited reserve of fossil fuel and negative consequences of conventional power sources lead to search more sustainable energy technology. Micro grid is an ensemble of Distributed Generation technologies, Energy Storage Systems and electrical and thermal loads which can operate autonomously or in grid connected mode [1] [2]. Its environmental friendly and it also contribute in local development by creating new jobs opportunities [3] [4].

Combined heat and power (CHP) systems, which are able to produce electricity and thermal energy from the same source, are a revolution in the world of power generation [1]. It allows reducing significantly the cost of production of energy to satisfy the electrical and thermal needs of rural isolated areas which have a low power demand and are not connected to the grid. Most of these areas have different types of energy resources to fulfill their need of energy.

In Morocco, firewood is used by 90% of rural households for their daily activities, namely cooking meals, baking bread and heating homes and water during winter [3]. However, the lack of sustainable management of the biomass resource leads to a national deficit of wood energy of 3.1 millions of tons per year and a loss of 31 Thousand hectares of wooded fields of which 75 percent is caused by firewood gathering [5].

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In this study a hybrid system based on wood chip CHP and solar PV was proposed to fulfill the power and heat needs of a rural village in the highs atlas mountains of morocco. The mean of this study is to improve the living conditions of the villagers by ensuring their electrical and thermal needs and reduce the consumption of wood to conserve the forest and reduce the national deficit of wood.

II. METHODOLOGY:

In this study the wood chip CHP is used to produce base power and heat demand of Imlil, the PV modules will be used to supply the peak load demand and the diesel generator as a backup when there is a deficit in the output of the renewable energy systems.

In order to have a correct analyze, all possible scenarios was represented to meet the load demand of the studied area. This study was treated as follow:

- Load profile analysis was carried out to know exactly the power and the heat demand of the village.
- Presentation of the village under study where its main characteristics, for instance climatic conditions, biomass potential and solar radiations data, were collected. These data were used afterward for the load assessment and for renewable energy production simulations.
- Technical specifications and economical details for the different sub-systems of the micro grid were given and used in Homer which allows us to compare many different design options based on their technical and economic merits and to give the optimal results.
- Multiple micro grid configurations were evaluated under the different load conditions and compared following their respective energy cost performance. The environmental impact of the different micro grid configurations is assessed as function of wood consumption.

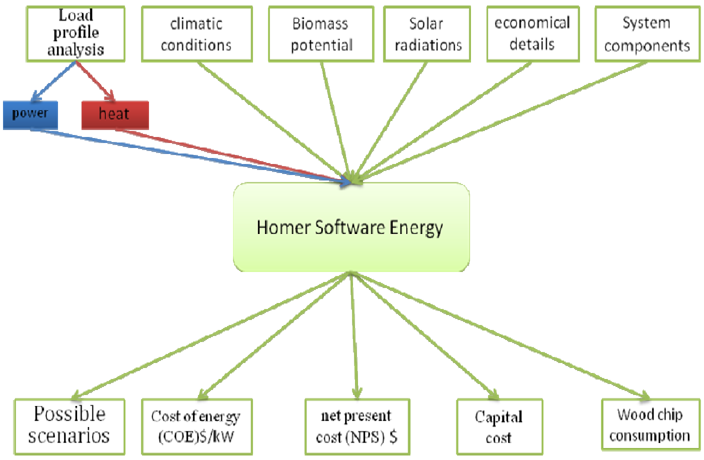

Fig. 1. explanatory diagram of the work methodology.

III. Load profile:

The energy demand of villagers are very limited compared with the townsfolk, the daily energy requirement of a villager does not exceed basic domestic uses such as cooking, lighting and heating in the winter, even the uses for fan as TV are very low.

Imlil is a Moroccan small village and in our energy estimate of power and heat we will consider a school, a mosque, a village office, five large houses and twenty small houses [3].

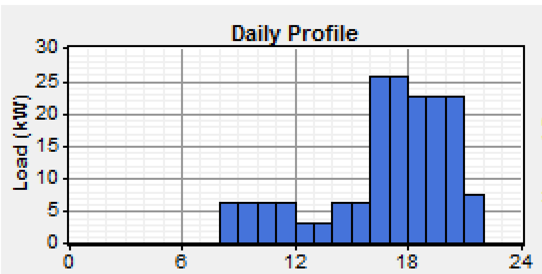

Fig. 2. Daily Electric profile.

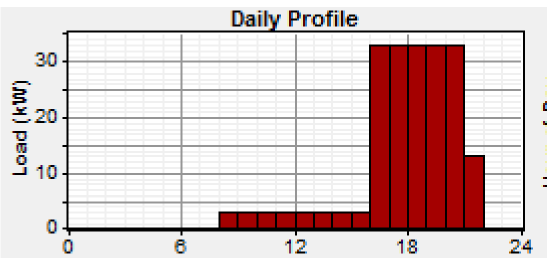

Fig. 3. Daily thermal profile.

IV. Studied village and Energy resources:

A. *Location and Climatic conditions of the Village:*

Imlil village is located in the center of Morocco in the High Atlas Mountains. Administratively, Imlil village belongs to the Azilal province in the Tadla-Azilal region. The village has a population of 9 796 inhabitants according to the census of 2014[6] with 1753 households. The latitude and longitude of Imlil are 31°03′43″North, 7°54′58″West respectively, with an elevation of about 3755 m [3].

The winter is cold and very wet on Imlil and its regions. TABLE I contains the monthly average maximum and minimum temperatures of Imlil. As can be seen the minimum is between -1 °C in January and 14 °C in July and August.

TABLE I. DIFFERENT END USERS COVERED BY THE CASE STUDY

Load	Summer (kw)			Winter (kw)
	Mai/Jun/ July	***Aug***	***Sept***	***Oct to Apr***
School	0	0	3,2	6,2
Small houses	15	15	15	35
Large houses	7,5	7,5	7,5	17,5
Mosque	2,6	2,6	2,6	5,6
Village Office	3	0	3	6
Total demand	28,1	28,1	31,3	70,3

TABLE II. MONTHLY AVERAGE MAXIMUM AND MINIMUM TEMPERATURE OF THE REGION OF IMLIL.

	Maximum[°C]	**Minimum[°C]**
Jan	8	-1
Feb	9	0
Mar	12	1
Apr	13	2
Mai	18	6
Jun	23	9
Jul	29	14
Aug	29	14
Sep	25	11
Oct	18	6
Nov	13	2
Dec	9	0

B. *Solar radiation:*

The annual average solar irradiation of the region is collected by Homer from the NASA Surface Meteorology and Solar Energy database. Fig. 4. shows that Imlil has a very important solar potential.

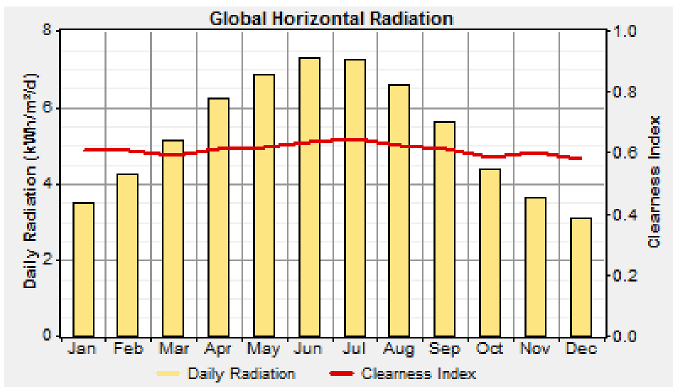

Fig. 4. Solar radiation of Imlil.

C. *Biomass resources of the region of IMLIL:*

The profile of the annual wood availability was taken from a study covering the Azzden valley zone [7] which is located close to the Imlil village. The annual consumption of wood data of Erdouz are taken to be representative for the case study carried out in this paper, and the results of a previous study [8] are used.

TABLE III. ESTIMATED BIOMASS IN [T/HA] OF DIFFERENT COMPARTMENTS OF THE THURIFEROUS GENÉVRIER VALLEY AZZADEN (HIGH ATLAS,MOROCCO).

Tree cover	Area [ha]	Plam	Troncs/ Branches	Leaves	Seed cones	Total Biomass
3-16%	744	0,57	24,85	1,11	0,07	27
16-26%	988	1,23	54,39	2,44	0,11	58
26-53%	236	1,58	62,09	3,14	0,17	67
53-100%	16	3,85	196,91	7,58	0,45	209

V. SYSTEM DESIGNATION AND ECONOMICAL DETAILS:

The basic hybrid system chosen in this study is made of a wood chip CHP system, PV modules, Batteries, converter and Diesel generator. All possible scenarios made by the different micro grid configurations of this hybrid system were optimized by the software to meet the village demand load with the most reasonable cost and the less consumption of wood.

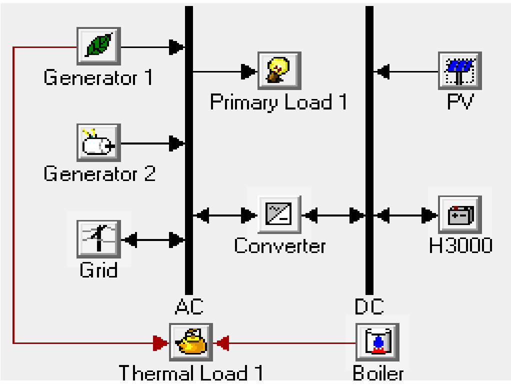

Fig. 5. Biomass/PV/Diesel/Battery hybrid system.

A. *Wood chip CHP:*

The CHP used in this study is the Froling's CHP fixed bed gasifier system his method of gasifying wood and subsequently extracting wood gas is a multistage, thermo chemical conversion process, similar to combustion.

Three CHP sizes are proposed to the Homer Energy System as shown in TABLE IV.

TABLE IV. TECHNICAL SPECIFICATIONS OF THE WOOD CHIPS CHP SYSTEM.

Technical specification		CHP		
Power consumption	kw	46	50	56
Thermal output	kw	95	105	115
Wood chip consumption	kg/h	35	37	40
Fuel heating output	kw	170	181	198
Thermal efficiency	%	56	58	58
Electrical efficiency	%	27	28	28
Overall efficiency	%	83	86	86

B. *Solar PV:*

PV system is employed in this study to meet the electrical power demand of the village, PV modules capital cost is 2000 $/kw with 2000 $/kw as replacement cost, 10 $/kw/y as O&M Cost and a lifetime of 25 years. All specifications are shown in TABLE V.

C. *Diesel generator:*

Diesel generators are usually employed to meet the peak load demand when there is a deficit in the output of the renewable energy systems, in this study it is employed as a backup for the power energy demand. Capital and replacement costs are boat 220 $/kw, O&M Cost is 160$/year/kw and lifetime is 15000h.

D. *Battery:*

The battery chosen in this study is Hoppecke 24 OPzS 3000 The nominal voltage of this battery is 2 V, the nominal capacity is 3000 Ah and the lifetime is 7 years. The capital and replacement costs are boat 1200 $/KW.

E. *Inverter:*

The converter was selected to be compatible with the PV array, converter cost is taken as $890/kW and replacement cost as $800/kW with a normal life of 15 years.

F. *Grid location:*

The grid is 15 Km away from Imlil, the grid connection lines cost is 176 687,02 $, Transformation station coast is 35 332,13 $ and the LV Distribution Grid is 11 777,15 $.

TABLE V. COMPONENTS RELATED COAST.

Options	Capital cost	Replacement Cost	O&M Cost	lifetime
CHP	300$/KW	300 $/KW	0,03/h	15000h
Solar PV	2000 $/KW	2000$/KW	1O$/kw/y	25 year
Battery	1200 $/Battery	1170 $/Battery	1O$/kw/y	7
Converter	890 $/KW	800 $/KW	1O$/kw/y	15
Diesel generator	220 $/KW	220 $/KW	0,03/h	15000h
Grid extensions	$190000/km	$190000/km	160$/km/y	

VI. TECHNICAL SOLUTIONS AVAILABLE AND ENERGY COST ANALYSIS:

The hybrid system chosen In this study is made of a wood chip CHP system Supplemented with PV modules, Batteries, converter and Diesel generator. Economic analysis is done by considering capital costs, replacement costs, O&M costs, and a lifetime of the components.

A. Wood chip CHP system:

In the first scenario the total of power and heat demand of the village will be assured only by a wood chip CHP system of 50 kW electric and 105 kw thermal. The Cost of energy (COE) and net present cost (NPS) are respectively 0,006 $/kwh and 62 742 $ with 3717 $/yr as operating cost of the system . This case is characterized by the lowest COE and Initial Capital of 15 225 $ due to the low cost of wood.

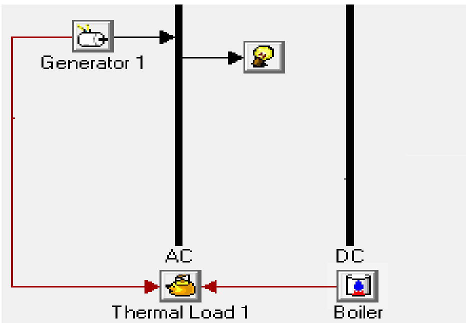

Fig. 6. Wood chips CHP system.

B. Wood chip CHP /Diesel generator hybrid system:

In this scenario also all heat and power demand was assured by the Wood chip CHP system of 50kw, however a diesel generator of 50KW was added as a backup, the COE of this case is 0,009 $/kwh which is a little bit higher than the first case, the NPS is 73173 $/kwh and the operating cost is 3307 $/yr.

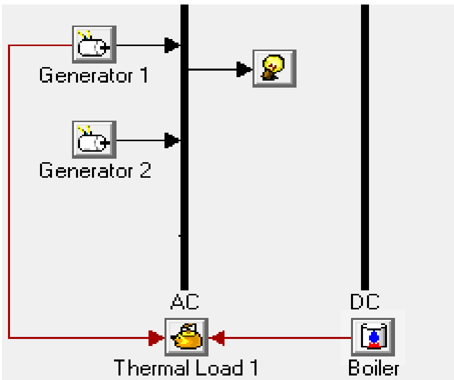

Fig. 7. Wood chip CHP /Diesel generator hybrid system.

C. Wood chip CHP /Battery hybrid system:

this system consist of only CHP system of 50 kw electric and 105 kw thermal combined with 10 stings of batteries in parallel which has a nominal capacity of 60 kwh and converters of 20kw capacity. it is very clear in (Fig. 12. & Fig. 13.) that this case has dumped the double of the previous systems capital cost and COE which are respectively 45025 $ and 0,019$/kwh, same for the NPC that has reached 105971$.

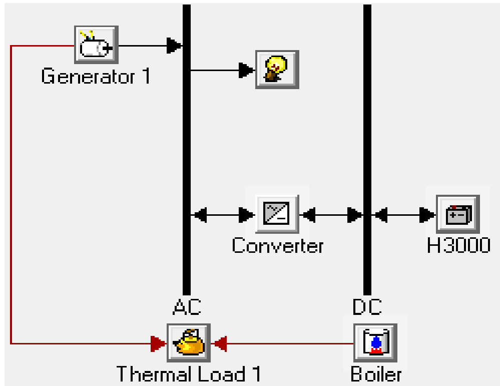

Fig. 8. Wood chip CHP /Battery hybrid system.

D. Wood chip CHP /PV/Battery hybrid system:

The fourth scenario includes wood chip CHP of 46 kw electric and 95 kw thermal, combined with 20 kw of PV modules and 10 stings of batteries in parallel, which has a nominal capacity of 60 kwh and converters of 20kw capacity. The NPS (119264 $) is slightly higher than the third case but the capital cost (65065 $) and the COE (0,025 $/Kwh) are much higher.

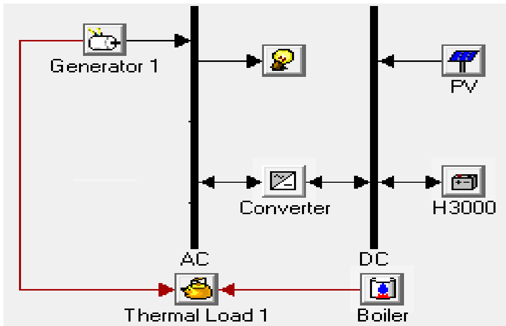

Fig. 9. Wood chip CHP /PV/Battery hybrid system.

E. Wood chip CHP/Diesel generator/ PV hybrid system:

The fifth scenario consist of wood chip CHP system of 46 Kw electric and 95 Kw thermal, combined with 20 Kw of PV modules, converters of 20kw capacity and diesel generator of 40KW as backup. this combination represent a capital cost(61025$), NPS(112256$) and COE (0,023$/Kwh) little bit lower than the fourth scenario.

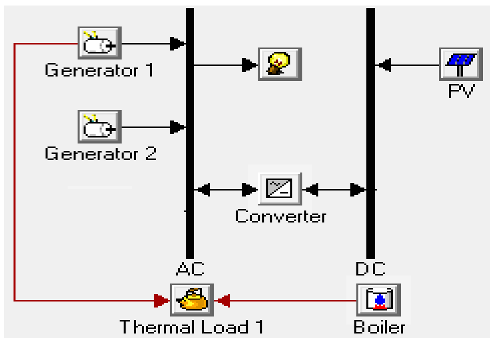

Fig. 10. Wood chip CHP/Diesel generator/ PV hybrid system.

F. *Wood chip CHP/Diesel generator/ PV/Battery hybrid system:*

This last hybrid renewable energy system consist of almost the same components of the fourth scenario, with a wood chip CHP of 46 kw electric and 95 kw thermal, 20 Kw of PV modules and 10 stings of batteries in parallel, which has a nominal capacity of 60 kwh ,converters of 20kw capacity and to give more stability to the system a diesel generator of 50 kw was added as backup. As shown in (Fig. 12. & Fig. 13.) this scenario is the most expensive of all renewable systems with a COE and NPS of 0,027 $/kwh and 125900$ respectively and 73075 $ as capital cost.

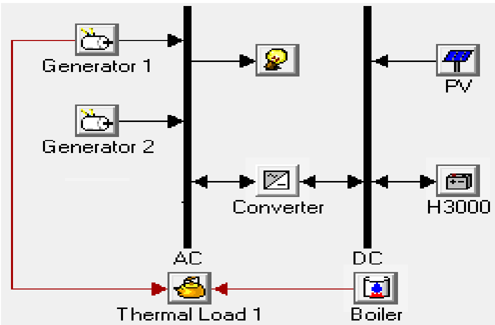

Fig. 11. Wood chip CHP/Diesel generator/ PV/Battery hybrid system.

G. *Grid extension:*

It is obviously shown (Fig. 12. & Fig. 13.) that this case represent the higher costs of all the suggested systems with a COE of 0,139 $/kwh and a NPS of 171740 $ which makes this case very excluded as a solution.

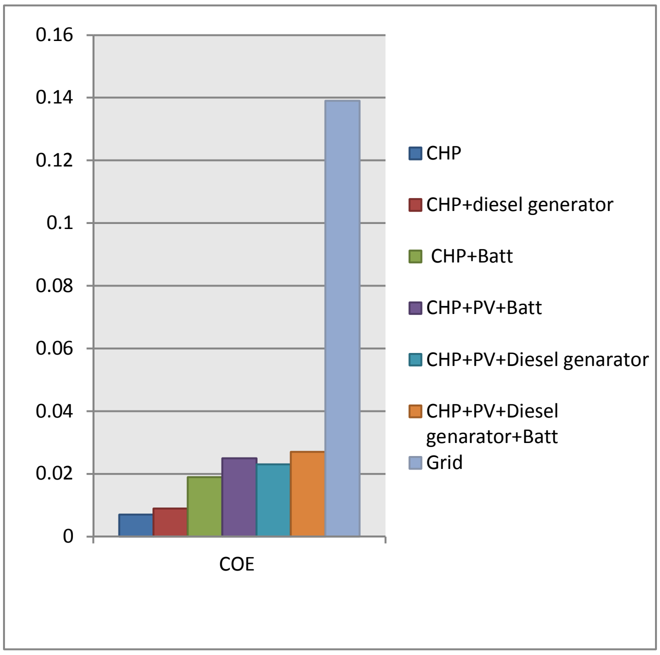

Fig. 12. The cost of electricity of all scenarios ($/kwh).

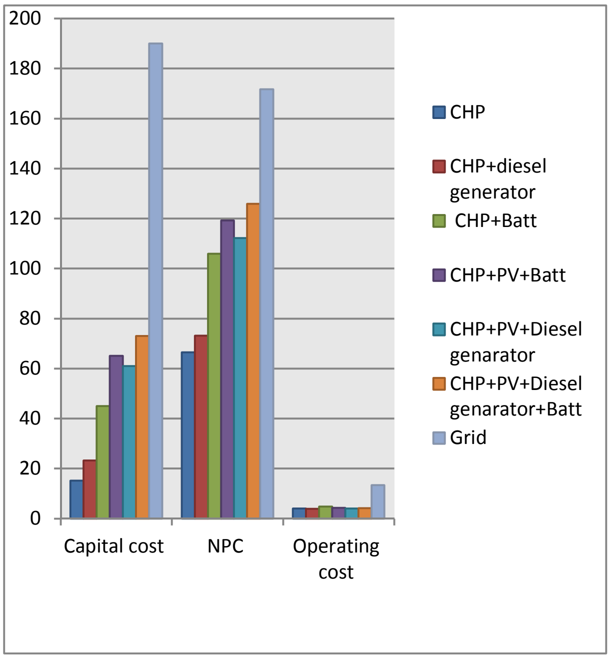

Fig. 13. NPC ($), Capital cost ($) and operating cost($/y). of all scenarios

VII. ENVIRONMENTAL BENEFIT:

A. *Wood consumption:*

Similarly to all Moroccan mountain villages, the villagers of Imlil use wood for cooking, and heating during the winter, to do this they turn to the random gathering of wood and use traditional ways of consumption which generates a very big waste of wood. The Moroccan state tries to fight against this random consumption and has launched several projects in this direction, notably the energy efficiency program in the sector of hammams and bakery ovens in Morocco [5].

The Renewable energy systems based on local resources in rural areas remains the most efficient way to electrify these villages, this approach will not only stop this bleeding of Moroccan forests but will also improve the living conditions of villagers, and will decrease the school dropout of the villager's children.

In our study it is very clear in (Fig. 12. & Fig. 13.) that we have been able to reach our goal in all the scenarios, the consumption of the wood of our systems did not affect the annual production of the village but on the contrary it remained slightly inferior in the three first cases and to score a very important difference in the last three which favors the use of supplementation by photovoltaic modules with the wood chips CHP system.

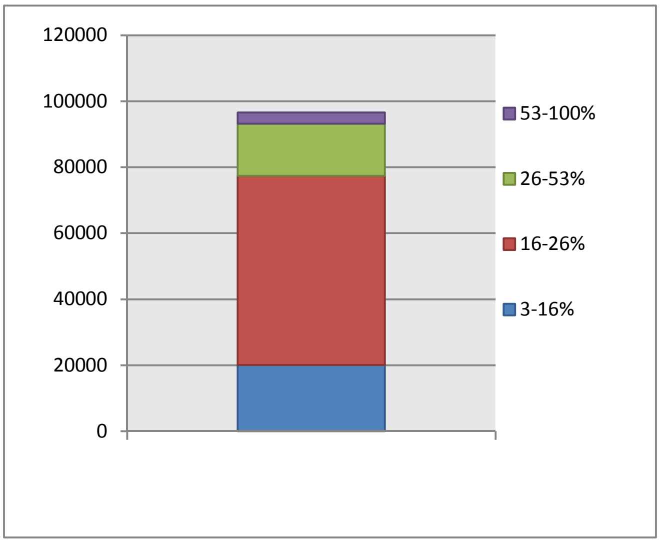

Fig. 14. Wood production of Erdouz t/y.

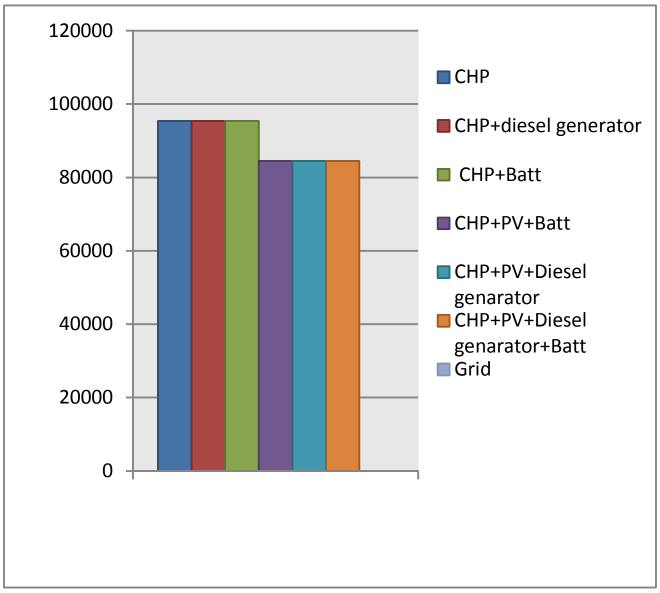

Fig. 15. Wood consumption of all proposed scenarios t/y.

VIII. CONCLUSION:

In this techno economic study a hybrid renewable systems was proposed to fulfill the energy need of Imlil. All reasonable combination of his components was represented to heat the power and heat demand with the most reasonable cost and the less consumption of wood. Total net present cost (NPC), cost of electricity (COE) and wood consumption were investigated.

This study has proved that grid extension is no more a good chose for powering rural eras and that the exploit of the local renewable energy potential of this village is the most economical feasible and environmental friendly solution.

The six scenarios represented in this study yielded highly resonant COE especially the first two cases. However, in terms of wood consumption, the last three scenarios, which include PV modules, showed a reduced consumption of wood compared to the other scenarios despite of the increase shown in COE, NPC and the capital cost.

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